

Reproducibility

Reproducibility in GNN is not an aspiration layered on top of the system; it is the operating contract that the pipeline enforces. The 0–24 processing steps are deterministic given a model specification and a target directory, and every published claim in this manuscript is traceable to a command that regenerates the underlying artifact. This section lists only commands that exist in the repository, so that a reader with a clean checkout can reproduce the pipeline, the validation gates, and this manuscript itself.

Pipeline Smoke Run I

The fastest way to confirm a working installation is to drive the full pipeline over the discrete model family without invoking the optional LLM steps:

```
uv run python src/main.py --target-dir input/gnn_files/discrete
```

This parses the discrete GNN files, runs visualization and rendering across the maintained backends, and writes all artifacts under the chosen output directory. The `--skip-llm` flag keeps the run hermetic and free of external API calls, which makes it suitable for continuous integration and for offline reproduction. To exercise the complete model corpus rather than a single family, point `--target-dir` at `input/gnn_files`, which contains 10 family directories.

GNN's reproducibility guarantees rest on a small set of strict, deterministic gates that bind the manuscript's quantitative claims to recomputable ledgers. The model-family acceptance gate runs the maintained families declared in the manifest and fails on any regression:

```
uv run python scripts/run_model_family_acceptance.py \  
    --manifest input/model_family_manifest.json --strict
```

The semantic-fidelity gate verifies that a parse $\rightarrow$ serialize $\rightarrow$ parse round trip preserves variables, edges, dimensions, parameter shapes, equations, time semantics, and ontology mappings across the 9 model families; the cross-framework gate profiles the 9 maintained backends and records explicit compatible and unsupported statuses rather than silently degrading. Both write their ledgers to an output directory of your choosing:

Validation Gates I

```
uv run python scripts/run_semantic_fidelity_gate.py \  
  --manifest input/model_family_manifest.json \  
  --output-dir output/semantic_fidelity --strict  
uv run python scripts/run_cross_framework_reliability.py \  
  --manifest input/model_family_manifest.json \  
  --output-dir output/cross_framework --strict
```

Validation Gates I

Under `--strict`, each gate exits non-zero on the first mismatch, so these commands double as assertions in an automated reproduction run. Code-quality reproducibility is enforced separately through the developer command reference: `just lint` runs the Ruff linter over `src` and `scripts`, and the broader `just quality` recipe chains formatting, terminology, documentation, type, and security checks for a full pre-commit gate.

This manuscript is itself a reproducible artifact. Every quantitative value in the prose — the pipeline step count, the family and backend counts, the source and test inventories — is a token rather than a hard-coded literal, and the deterministic producer regenerates all of them from the live repository state:

```
python scripts/z_generate_manuscript_variables.py
```

That command recomputes the `{{...}}` tokens, persists them to `output/data/manuscript_variables.json` for audit, and hydrates the manuscript sources into `output/manuscript/`. The hydrated sources are then rendered to PDF through the template research pipeline (invoked from the parent monorepo / template framework), whose rendering stage converts the markdown sections, figures, and references into the final document:

```
python -m scripts.manuscript_build_figures  
python -m scripts.z_generate_manuscript_variables
```

Because the variables file is regenerated immediately before rendering, the numbers in the rendered PDF cannot drift from the repository: a code change that alters, for example, the test inventory (365 test files, 4109 test functions) propagates into the prose on the next regeneration without any manual editing.

Reproducibility Contract I

Reproducibility Contract I

- ▶ Do not cite results that cannot be regenerated or directly traced to a command in this repository.
- ▶ Keep generated outputs under `output/` and maintained manuscript source under `manuscript/`; treat everything in `output/` as disposable and regeneratable.
- ▶ Express every quantitative claim in the prose as a double-brace `{{...}}` token substituted by `scripts/z_generate_manuscript_variables.py`, never as a hard-coded number.
- ▶ Keep private data, credentials, and unpublished sensitive details out of the manuscript and out of version control.
- ▶ Record the exact verification commands — the smoke run, the acceptance and fidelity gates, and just `lint` — before marking this manuscript publication-ready.